Group video project

During week 14, you will work with your group to make a video explaining an abstract algebra topic of your choice (probably related to at least one member's blog post). During week 15 we will watch these videos and host a "Question and Answer" session after each video.

Goals

- Teach your classmates about some interesting ideas in Abstract Algebra!
- Get more comfortable using "imperfect takes."
- Learn how to make and upload a YouTube video.
- Learn the basics of video editing software.

Guidelines

1. You have license to make any style of video that you would like, ranging from a Numberphile-style video to a series of short-format TikTok videos.
2. Your video (or series of videos) should be 10 minutes in length, with a strict requirement to be 8–15 minutes total.
3. Your video should contain every member of your group appearing either in the video or as a video voice-over.
4. Follow each of the steps listed in the checklists below.

Checklists

Rules like these are allowed to be broken, but make sure you have a good reason to break them!

Getting started checklist

- Watch some examples both from the Video Inspiration section below and student videos posted on Discord.
- Watch at least 2.5 minutes of the Grant Sanderson advice video embedded below, starting at 8:32.
- Devise a rough outline for a 8–15 minute video aimed at your classmates.

Pre-recording checklist

- Find a time for your group to get together to film together. (No filming on your own!)
- Gather two or more cameras with microphones (screen capture can count here).
- Plan to film actual handwriting on a surface that is not a chalkboard or whiteboard. (No videos that are just slide shows.)

Editing checklist

- Include at least two "camera angles."
- Edit audio to ensure that volume is consistent.
- Include an image or external video with audio. (An image from the blog post could be a good choice here.)
- Ensure that the video is between 8–15 minutes.

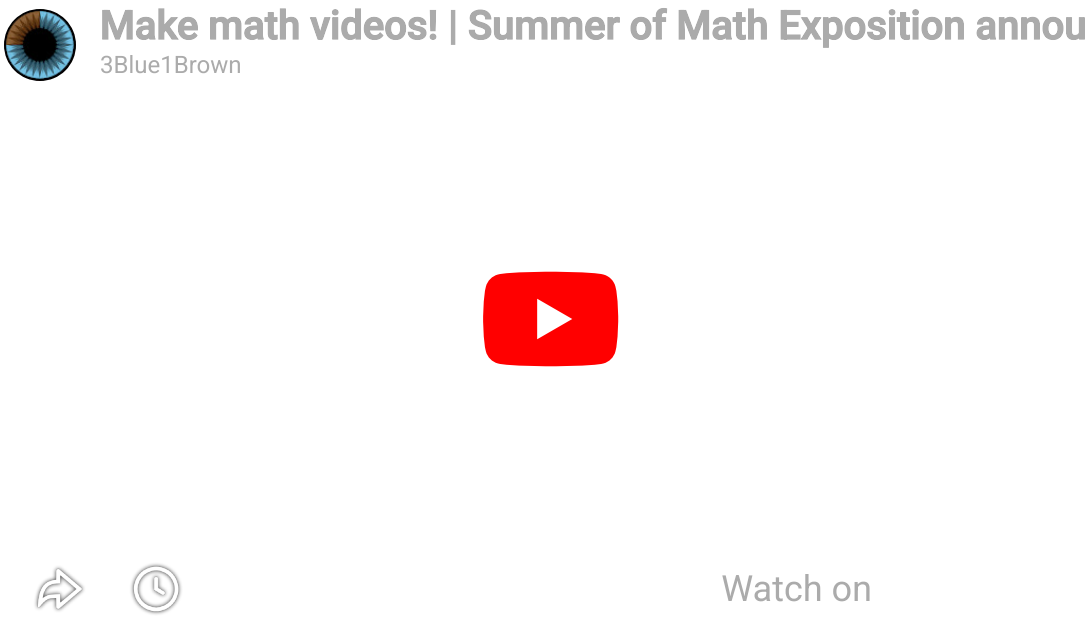
Wrapping up checklist

- Upload the video to a video service such as YouTube (can be unlisted).
- Ensure that the video title is interesting, catchy, and descriptive.
- Include a meaningful video description.
- Spend a few minutes trying to anticipate some questions you might get about the video when you play it in class on Monday, May 4th.
- Turn it in on Gradescope by providing a PDF that simply contains a link to the video! Group submissions are turned on. If you use the `\usepackage{hyperref}` at the top of your LaTeX file, then you can include the URL with `\url{...}`.

Best practices

- Ensure that your videos are as example-driven as possible! In particular, motivate any definitions that you use! Here's a video by Grant Sanderson of 3Blue1Brown

that expands on this (watch from at least 8:32–11:02).



- Watch the student videos I've posted to Discord along with some of the following videos:

Video inspiration

Watch some of the student videos posted to Discord as well as some of the following YouTube videos.

Numberphile

- The Plotting of Beautiful Curves (Euler Spirals and Sierpiński Triangles)
- Hitomezashi Stitch Patterns

GrapefruitGecko

- Understanding Modular Multiplication Table
- The Gauss-Bonnet theorem | Folding up Surfaces (Good use of a whiteboard!)
- A cozy history of hyperbolic geometry

Mathologer

- Euler's infinite pi formula generator

3Blue1Brown

- Why this puzzle is impossible

Summer of Math Exposition (SOME)

- SOME playlist

Miscellaneous

- You Could Have Invented Homology, Boarbarktree (content, not animation)
- Crocheting Hyperbolic Planes, Daina Taimiņa at TEDxRiga

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